Project Proposal

1 Project Title

Investigating the Effect of Acoustic Feature Representation on Musical Instrument Recognition

2 Student Information

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3 Project Overview

Musical instruments produce distinctive sounds due to differences in their spectral, temporal, and harmonic characteristics. Automatic musical instrument recognition aims to identify the instrument producing a given audio signal and has applications in music information retrieval, automatic music transcription, and audio analysis. However, different instruments may share similar acoustic characteristics, and recognition performance can depend strongly on how the audio signal is represented.

This project investigates which acoustic features are most informative for distinguishing musical instruments and how different feature representations affect classification performance. Three groups of features will be compared: Mel-frequency cepstral coefficients (MFCCs), hand-crafted spectral features such as spectral centroid, bandwidth, roll-off and flux, and temporal features such as RMS energy, zero-crossing rate, and attack characteristics.

NSynth will be used as the primary dataset because it provides labelled monophonic instrument notes and instrument-family annotations [1, 2]. IRMAS and/or Medley-solos-DB may be used for external validation if time permits [3–6]. Feature combinations will also be investigated to determine whether complementary acoustic information improves recognition performance. Accuracy, macro F1-score, and confusion matrices will be used for quantitative evaluation.

4 Background and Motivation

Musical instrument recognition has been a longstanding problem in music information retrieval, with a central challenge being the computational modelling of timbre—the perceptual quality that allows listeners to distinguish instruments even when they produce similar pitches and loudness levels. Saldanha and Corso demonstrated that listeners could identify instruments from acoustic characteristics, while also observing differences in identification accuracy between instruments [7]. Grey later investigated the perceptual structure of timbre and showed that it involves multiple dimensions, with spectral energy distribution and temporal characteristics of the attack playing important roles [8]. These findings motivate the investigation of both frequency-domain and time-domain characteristics for instrument recognition.

MFCCs are widely used in audio classification because they provide a compact representation of the spectral envelope using an auditory-inspired frequency scale. Other hand-crafted features, including spectral centroid, bandwidth, roll-off, and spectral flux, have also been investigated for modelling instrument timbre. Nielsen et al. examined the relevance of spectral features for musical instrument classification, demonstrating the importance of spectral characteristics in instrument modelling [9]. More recent work has investigated feature combinations. More recent work has also investigated feature combinations. Liu et al. reported strong performance on Medley-solos-DB using multi-channel feature fusion with XGBoost [10], demonstrating the potential benefit of combining complementary acoustic information.

These studies suggest that different feature representations capture different aspects of instrument sounds. This project therefore focuses on a controlled comparison of MFCC-based, spectral, and temporal representations, examining both individual and combined feature groups and whether their effectiveness varies across instrument families.

5 Proposed Methodology

5.1 Dataset and Data Preparation

NSynth will be the primary dataset. It contains 305,979 four-second monophonic notes sampled at 16 kHz from 1,006 instruments, with annotations including instrument family [1,2]. A balanced subset of instrument classes will be selected to maintain computational feasibility while representing different instrument families.

IRMAS and/or Medley-solos-DB may be used as external validation datasets. They will be evaluated separately rather than combined with NSynth because they have different recording characteristics and label taxonomies [3–6].

Audio will be processed using a consistent pipeline, including sampling-rate conversion where necessary, normalisation, and fixed-length segmentation. Training, validation, and test sets will be separated before model training to reduce information leakage.

5.2 Acoustic Feature Extraction

Three feature groups will be investigated.

MFCC-based representation: Thirteen MFCC coefficients and their Delta and Delta-Delta coefficients will be extracted from the STFT using a Mel-scaled filterbank. MFCCs provide a compact representation of the spectral envelope and have been investigated for musical instrument modelling [9].

Spectral representation: Spectral centroid, bandwidth, roll-off, flatness, and flux will be extracted from the STFT to describe spectral energy distribution and variation.

Temporal representation: RMS energy, zero-crossing rate, and statistics describing the temporal envelope and attack characteristics will be extracted to capture onset and decay behaviour.

Frame-level measurements will be aggregated using statistics such as mean and standard deviation to produce fixed-dimensional representations for classification.

5.3 Classification and Experimental Design

A Support Vector Machine (SVM) will be used as the main classifier. The same classifier configuration, data split, and evaluation procedure will be used across feature groups to provide a controlled comparison.

Five experiments will be conducted:

1. **Baseline:** MFCC-based SVM classification.

2. **Feature Comparison:** Separate evaluation of MFCC, spectral, and temporal representations.
3. **Feature Combination:** Comparison of MFCC + spectral, MFCC + temporal, and MFCC + spectral + temporal representations.
4. **Instrument-Family Analysis:** Comparison of feature performance across different instrument families to investigate whether particular groups depend more strongly on spectral or temporal information.
5. **Acoustic Error Analysis:** Examination of confusion matrices and representative errors using spectra, spectrograms, harmonic structure, and temporal envelopes.

5.4 Tools and Evaluation

The project will be implemented primarily in Python. Librosa, NumPy, and SciPy will be used for audio processing and feature extraction, while scikit-learn will be used for SVM classification and evaluation. Matplotlib will be used for visualisation.

Accuracy and macro-averaged F1-score will be the primary evaluation metrics. Confusion matrices will be used to identify systematic classification errors. The experiments will use consistent data splits and classifier settings to ensure meaningful comparisons.

6 Expected Outcomes

The project is expected to deliver a working Python-based musical instrument recognition prototype supporting MFCC-based, spectral, temporal, and combined feature representations.

The main outcome will be a quantitative comparison of the feature groups, identifying which representations provide the most useful information and whether combining complementary features improves classification performance. Family-wise analysis will investigate whether the relative importance of spectral and temporal information varies across instrument groups.

Accuracy and macro-averaged F1-score will be the primary evaluation metrics, while confusion matrices will be used to identify systematic classification errors. The project will also provide acoustic analysis of representative classification errors using spectrograms, spectra, harmonic structure, and temporal envelopes.

The final GitHub repository will contain the source code, experimental configurations, results, visualisations, documentation, and a demonstration of the recognition system.

7 Timeline

Week	Task
1–2	Finalise project topic and research questions.
3–5	Literature review; finalise dataset and data preprocessing pipeline.
6–9	Implement MFCC, spectral and temporal feature extraction; develop baseline SVM and conduct feature comparison experiments.
10–11	Feature combination experiments; instrument-family analysis; optimisation and evaluation.
12–13	Finalise results and analysis; complete final report and GitHub documentation.

References

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